

Series and infinite products – 3

Conditional convergence, rearrangements and signs

Homework

12 сентября 2026 года

Theoretical minimum. Conditional convergence means $\sum a_n^+ = \sum a_n^- = \infty$ with $a_n \rightarrow 0$, and this is exactly what makes Riemann's rearrangement construction work. Absolute convergence is invariant under all permutations; conditional convergence survives permutations of bounded displacement. In $\mathbb{R}^d$ the set of sums obtainable by rearrangement is an affine subspace (Lévy–Steinitz). Brackets may be inserted freely and removed only when the terms tend to zero and the groups are of bounded length. For signs there are two different thresholds: some choice of signs works whenever $a_n \downarrow 0$, all choices work only for absolutely convergent series, and random signs work exactly when $\sum a_n^2 < \infty$. Finally, the set of subsums of a series with positive terms is either an interval, or a Cantor set, or a finite union of intervals — nothing else.

1. (IMC 2010) Let $\mathbb{N}$ be the set of all positive integers.
 - a) Is it true that for every bijection $f: \mathbb{N} \rightarrow \mathbb{N}$ the series $\sum_{n=1}^{\infty} \frac{1}{nf(n)}$ is convergent?
 - b) Prove that there exists a bijection $f: \mathbb{N} \rightarrow \mathbb{N}$ such that the series $\sum_{n=1}^{\infty} \frac{1}{n+f(n)}$ is convergent.
2. (IMC 2013) Does there exist a sequence $(a_n)_{n \geq 1}$ of complex numbers such that for every positive integer p we have that $\sum_{n=1}^{\infty} a_n^p$ converges if and only if p is not a prime?
3. Prove that a series $\sum a_n$ converges absolutely if and only if every rearrangement of it converges, and if and only if every subseries converges.

4. Prove the easy half of the Lévy–Steinitz theorem: the set of sums of a convergent series in $\mathbb{R}^d$ is closed under the operations that make it an affine subspace, and exhibit a series in $\mathbb{R}^2$ whose set of sums is exactly a line.
5. Prove that brackets of bounded length can be removed when $a_n \rightarrow 0$, and give an example of a divergent series which becomes convergent after bracketing in blocks of length two, and another one where the blocks must have unbounded length.
6. (Putnam 2020) Let n be a positive integer. Prove that $\sum_{k=1}^n (-1)^{\lfloor k(\sqrt{2}-1) \rfloor} \geq 0$.
7. Let $\sum a_n$ converge conditionally. Prove that the partial sums of the rearranged series can be made to oscillate between any two prescribed numbers $\alpha < \beta$, and that the set of limit points of the partial sums can be made to be any prescribed closed interval.
8. Prove that if $\sum a_n^2 < \infty$, then for almost every choice of independent random signs $\varepsilon_n = \pm 1$ the series $\sum \varepsilon_n a_n$ converges. (Estimate the variance of the partial sums and use the martingale or the second-moment argument of Probability – 1.) Show that the condition is sharp.
9. (OMOUS 2023) The series $\sum_{n \geq 1} x_n$ is

$$1 - 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \cdots + \underbrace{\frac{1}{2^n} - \frac{1}{2^n} + \cdots + \frac{1}{2^n} - \frac{1}{2^n}}_{2^{n+1} \text{ summands}} + \cdots$$

- a) Find its sum.
 - b) For the bijection $\pi: \mathbb{N} \rightarrow \mathbb{N}$ given by $\pi(3k) = 2k$, $\pi(3k-1) = 4k-1$, $\pi(3k-2) = 4k-3$ find $\sum_{n \geq 1} x_{\pi(n)}$.
 - c) Construct a bijection $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ with $\sum_{n \geq 1} x_{\sigma(n)} = 2$.
10. Let σ be a permutation of $\mathbb{N}$ such that for every conditionally convergent series $\sum a_n$ the rearranged series $\sum a_{\sigma(n)}$ converges to the same sum. Prove that the number of maximal intervals of integers in $\sigma(\{1, \dots, n\})$ is bounded, and that this condition is also sufficient.